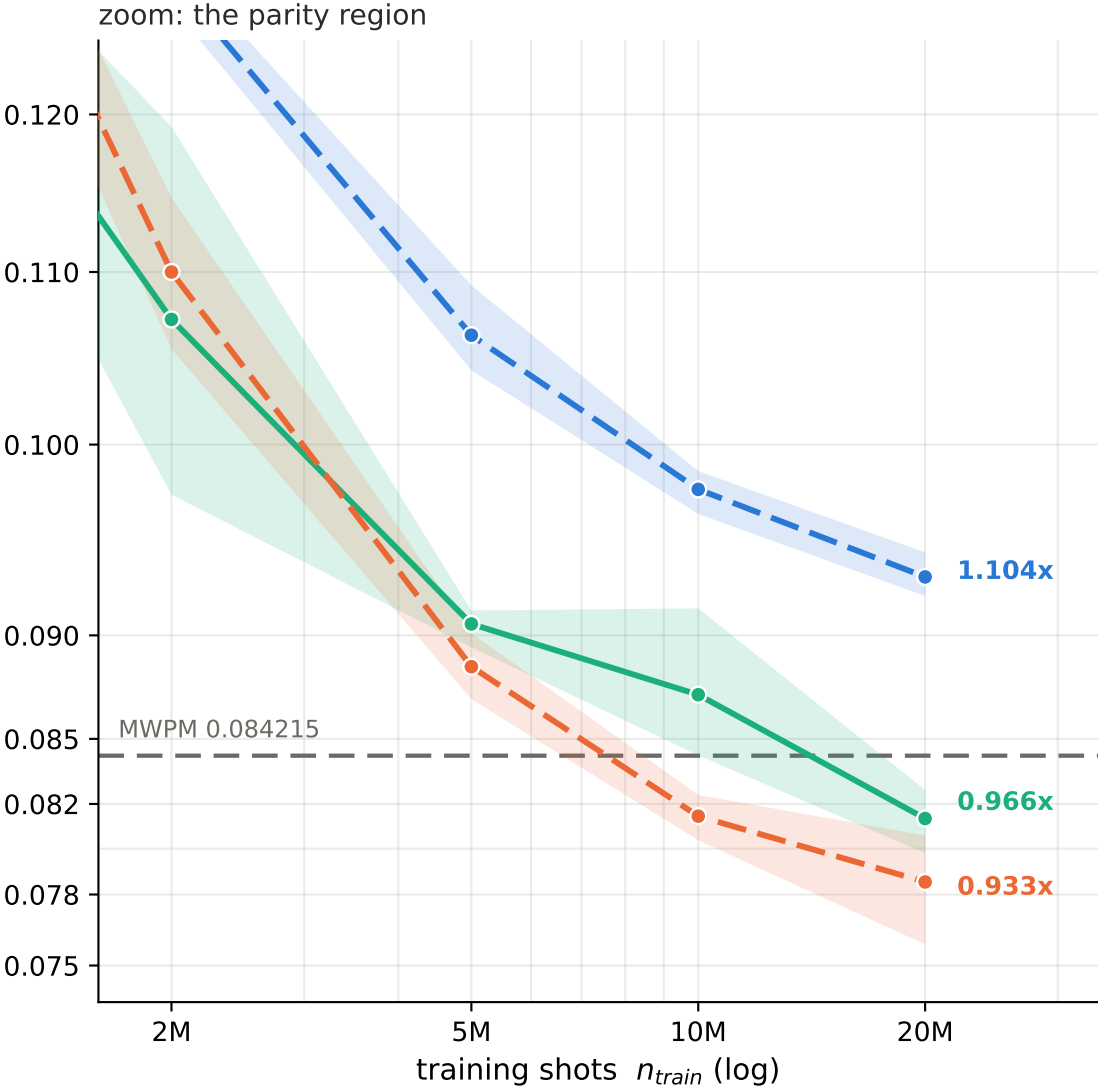
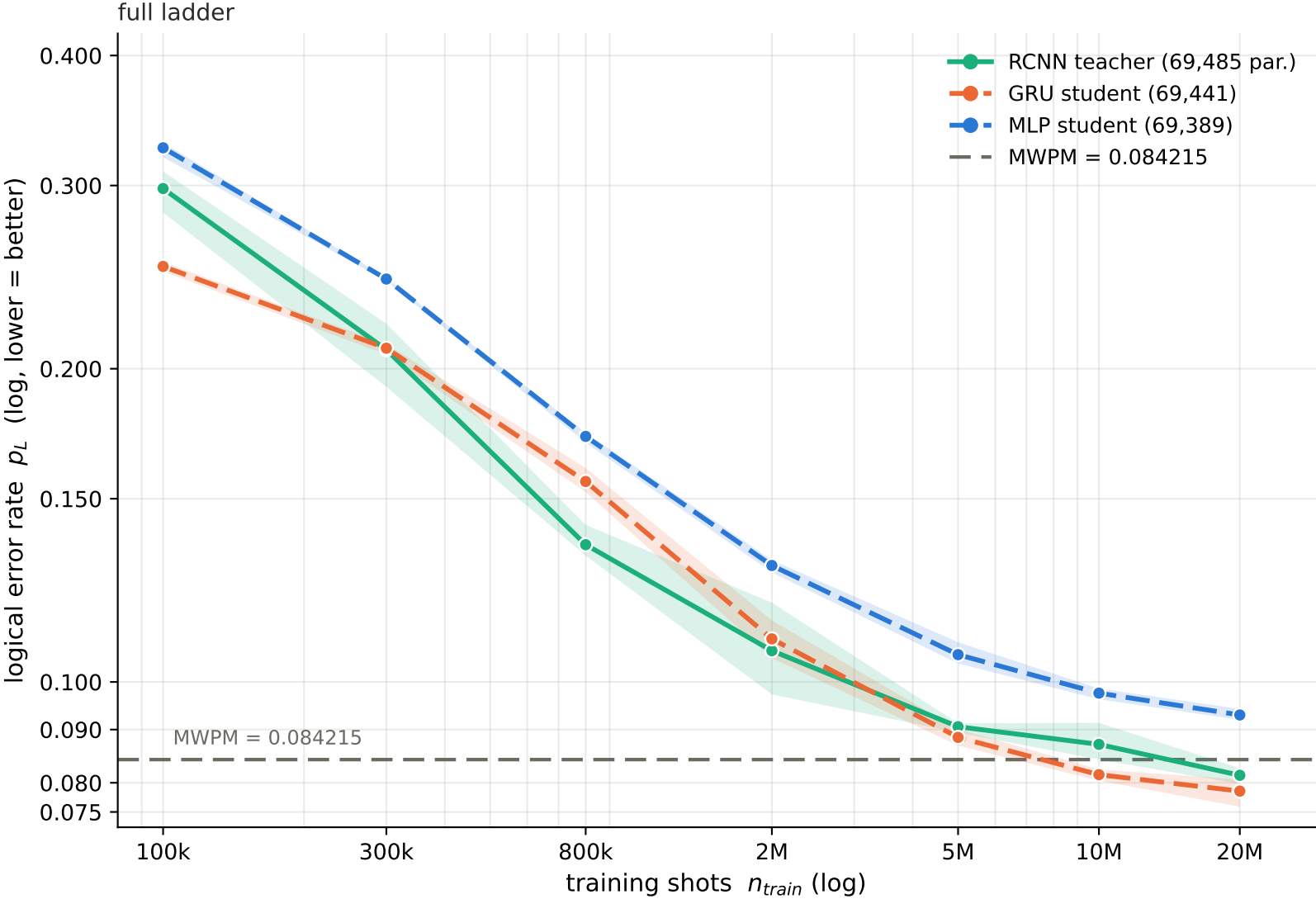


Experiment 8 — RCNN teacher vs hls4ml-native students at $r=5$

Surface code $d=5$, $p=0.01$, $r=5$ · 200,000-shot validation partition [20.0M, 20.2M) · 3 seeds per rung (teacher 20M: 1) · band = full range



Matched capacity: RCNN 69,485 parameters, GRU 69,441, MLP 69,389. Students are hard-label trained ($\alpha=1$), NOT distilled — the distilled arms need an $r=5$ teacher cache. p_L is each run's re-scored .best checkpoint on the validation partition; MWPM is recomputed there too (the 0.08910 in the training logs belongs to the old $r=5$ pool's 10k tail).